Neural Networks: Course Introduction

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Neural Networks 2026/2027

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SAPIENZA
UNIVERSITÀ DI ROMA

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Your teachers





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General information

- Credits: 6 CFU
- Course language: English
- Offered programs: Master's Degrees in Artificial Intelligence and Robotics, Engineering in Computer Science, optional for any course of study
- Course webpage: https://sites.google.com/uniroma1.it/neuralnetworks2026/
- Classroom page: Download slides, notebooks, and additional material. Also used for news and important communications. Access code: **sy3uqdop** . Please register here.
- Offices: DIET Dept. (SPV), RM032, 1st floor
- Office hours: by appointment
- Laboratory: ISPAMM Lab, DIET Dept.

ISPAMM Research Lab



Figure 1: The **Intelligent Signal Processing and MultiMedia (ISPAMM)** research team (Website: www.ispamm.it).

Course description

The course is intended as a broad overview to neural networks, as used today in a number of applicative fields.

It provides both theoretical and practical understanding of how neural networks are designed and implemented.

We explore how to address *machine learning* and *deep learning* tasks with neural networks.

We introduce essential components to deal with images (*convolutive layers*), sequences (*recurrent layers*), and sets (*transformer layers*).

We finally focus on a selection of advanced and emerging topics, including *multimodal neural networks* and *explainability*, among others.

Expected results

At the end of the course, a student is expected to have:

- a broad understanding of how neural networks work in practice,
- the capability of implementing new components from scratch,
- the capability of re-using existing models,
- the capability of designing new architectures for problems beyond the overview of the course,
- the ability to autonomously study new topics on the research frontier, and navigate the current scientific literature and software panorama.

What we expect you to do

- Be curious and passionate about AI and Neural Networks.
- Ask yourself what your main interests in Neural Networks are.
- Consider the practical part of this course as an opportunity for you to understand your attitudes.
- Attend classes, as it is the first step to become an AI master!

Prerequisites and background

This course requires basic knowledge of:

- machine learning fundamentals
- linear algebra
- probability theory and stochastic processes
- optimization
- Python

Anyway, sufficient recalls will be given and additional material can be provided, even on request, to fill some gaps.

Syllabus I

- **Introduction and Background**
  - Introduction to neural networks
  - Background recalls on linear algebra, probability and optimization, computation
- **Neural Networks Fundamentals**
  - Linear regression, training and automatic differentiation
  - Multi-layer perceptrons
- **Core Neural Network Models**
  - Convolutional neural networks and modern architectures
  - Recurrent neural networks and modern architectures
  - Transformers and attention-based architectures
  - Graph neural networks

Syllabus II

- **Efficient Neural Networks**
  - Quantization and pruning
  - Dynamic sparsity
  - Hypercomplex neural networks
- **Multimodal Neural Networks**
  - Multimodal representation learning
  - Joint embedding predictive architectures
  - Large-scale multimodal networks
- **Explainability and Interpretability**
  - Explainability techniques and case studies

Course material

General information:

- On the NN course page you can find all the course information, including program and exam dates.

Main material:

- Lecture slides, papers, notebooks, supplementary material, and news will be provided on the Classroom page (access code: 7uynqjmd).

Main textbooks (available online):

- Simone Scardapane, “Alice’s Adventures in a Differentiable Wonderland", 2024.
- Simon J. D. Prince, “Understanding Deep Learning”, MIT Press, 2023.
- Aston Zhang, Zachary C. Lipton, Mu Li, and Alexander J. Smola, “Dive Into Deep Learning”, 2023.

Class schedule

The course starts on September 25, 2026 and will end on December 18, 2026, with the following schedule:

- Tuesday, 13:30–16:00, Classroom A3, Via Ariosto, 25
- Friday, 12:00–13:30, Classroom A5, Via Ariosto, 25

The course is held in in-presence modality only.

Exam assessment and grade evaluation

- **Default option**
  - Oral examination on the course program (usually 2 topics).
- **Research option**
  - Research projects on emerging and advanced topics (proposed by professors only).
  - It could be useful to run for the lode or to prepare for a possible follow-up thesis.

Exam schedule

- **I Session:** January TBD, 2027
- **II Session:** February TBD, 2027
- **I Extraordinary Session:** March TBD, 2027
- **III Session:** June TBD, 2027
- **IV Session:** July TBD, 2027
- **V Session:** September TBD, 2027
- **II Extraordinary Session:** October TBD, 2027

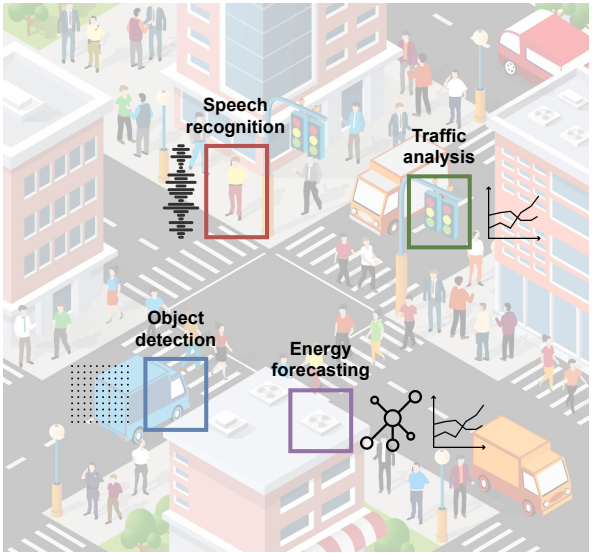
2 What are neural networks?

Neural Networks Daily
Definition of Neural Networks
Historical Evolution of Neural Networks
Neural Networks vs Machine Learning

Neural networks are everywhere



Neural networks are everywhere



Neural networks daily

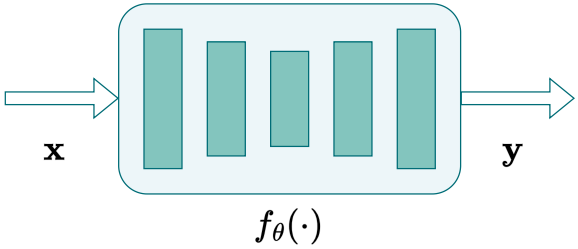
Neural networks power many AI applications we use every day, often without realizing it.

Examples of AI systems relying on neural networks, among others:

- Voice Assistants: Siri, Alexa, etc., use speech recognition systems.
- Image Tagging: Social media platforms automatically tag faces in photos.
- Recommendation Systems: Streaming services suggest movies and music.
- Autonomous Driving: Modern cars use neural networks for object detection and navigation.

What is a neural network?

Neural networks are **composable**, **differentiable**, and **parameterized functions** that can be optimized end-to-end using gradient-based methods.



They consist of interconnected layers (*processing blocks*) that extract hierarchical features from complex, high-dimensional data.

This design enables them to learn intricate patterns and representations from raw inputs.

Neural networks as processing blocks of complex data

A neural network functions as a processing block that converts complex inputs into meaningful outputs.

Data flow:

1. *Input:* Raw data (e.g., images, text, audio).
2. *Hidden Processing:* Sequential layers extract and transform features.
3. *Output:* Final predictions (e.g., class labels, regression values).

This modular scheme simplifies handling and understanding complex, high-dimensional data.

Modular structure of neural networks

Neural networks are built from simple computational blocks called neurons.

These neurons are organized into layers:

- Input Layer: Receives raw data.
- Hidden Layers: Process and extract features.
- Output Layer: Produces the final prediction or classification.

Each layer applies a simple transformation; their composition yields a powerful model.

Key components of a neural network

Neurons: The basic units that perform simple computations.

Weights & biases: Parameters that scale and shift the input signals.

Activation functions: Introduce non-linearity to enable complex decision boundaries.

Layers: Groups of neurons that work together to process data.

Learnable parameters

All inputs and outputs are represented as tensors—large, n-dimensional arrays of numbers.

Neural networks consist of multiple blocks (layers), each performing a simple operation on these tensors.

Each layer uses parameters (e.g., in a matrix multiplication) that adjust the input data. These parameters are learned during training.

The entire network is optimized end-to-end via numerical methods, improving its performance on a given dataset.

Differentiability and end-to-end optimization

The differentiable nature of neural networks allows the use of gradient-based optimization techniques.

End-to-end training: All parameters are adjusted simultaneously to minimize a *loss function* over a training dataset.

This smooth, numerical optimization is key to the network's ability to learn from data.

Historical evolution of neural networks

- **1950s:** Early work with perceptrons laid the groundwork.
- **1980s:** The introduction of backpropagation enabled multi-layer networks.
- **2010s:** Deep architectures (many layers) have revolutionized fields like *computer vision*.
- **2020s:** Advanced models are now pushing the boundaries in natural language processing, reasoning, and multimodal integration.

Neural networks vs machine learning & deep learning

Machine Learning: The broad field of algorithms that enable computers to learn from data. It includes models and learning paradigms.

Deep Learning: It is an extension of machine learning when the amount of data is significantly big to require large-scale models and different paradigms capable of learning highly abstract representations.

Neural Networks: A class of models that due to their capabilities and modulatities can be used from machine learning to deep learning problems.

Neural networks vs machine learning & deep learning



Figure 2: Neural networks are a class of models that can be designed for different levels of complexity, from machine learning to deep learning problems.

Summary and Further Reading

Neural networks are composable, differentiable models that process complex data via interconnected layers.

They represent a key approach within machine learning and are at the heart of modern deep learning.

For more in-depth study, consider resources such as:

- Simone Scardapane, “Alice’s Adventures in a Differentiable Wonderland", 2024.
- Simon J. D. Prince, “Understanding Deep Learning”, MIT Press, 2023.
- Aston Zhang, Zachary C. Lipton, Mu Li, and Alexander J. Smola, “Dive Into Deep Learning”, 2023.

Next lecture

- The mathematics of neural networks
  - The notions of linear algebra that can be considered most relevant to neural networks will be introduced.
  - Basic operations will be shown to process and manipulate structured data.

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https://sites.google.com/uniroma1.it/neuralnetworks2026

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